A Unified CT and MR Enterography Foundation-Model Pipeline for Individualized Prognosis in Crohn's Disease

Ben Shaya
Moti Freiman[1]

Abstract

Crohn's disease (CD) is assessed with both magnetic resonance enterography (MRE) and computed tomography enterography (CTE), yet deep-learning pipelines are almost always trained on a single modality, are annotated by hand at small scale, and stop at cross-sectional activity scoring rather than individualized prognosis. We assembled 8,053 enterography studies from four centers and built a two-stage pipeline in which a clinical language model (HSMP-BERT) automatically extracts ten radiological findings from free-text Hebrew reports, a modality-conditioned foundation-model backbone (frozen DINOv3 with low-rank adapters and dual-branch multi-instance learning) detects those findings from both CTE and MRE, and a penalized Cox model maps findings and clinical features to time-to-event risk. After excluding ulcerative colitis to obtain a rigorously CD-only cohort (7,170 detection studies; 5,148 with linked outcomes), the unified detector reached a held-out macro-AUC of 0.780 (95% CI 0.750–0.810) on MRE and 0.744 (0.691–0.789) on CTE, and predicted time to surgery with a concordance index of 0.792 (0.776–0.808), where imaging added 0.103 over clinical features alone, generalized across centers (leave-one-center-out 0.754), and was well calibrated (integrated Brier score 0.074). A modality token with feature-wise linear modulation let one unified model match the MRI specialist (0.780 vs 0.793) and outperform the CT specialist (0.744 vs 0.673), whereas domain-specific normalization degraded the dominant modality. A single conditioning-light foundation model can therefore serve both modalities and deliver individualized, calibrated CD prognosis.

Introduction

Crohn's disease (CD) is a chronic relapsing–remitting inflammatory bowel disease (IBD) affecting over three million people in the United States, with rising global incidence . Its clinical course is notoriously heterogeneous: some patients maintain long-term remission with minimal therapy, while others progress to surgical resection, steroid-dependent disease, or repeated escalation of biologic therapy . This unpredictability poses a fundamental clinical challenge, namely which patients will deteriorate, and when.

Cross-sectional enterography is central to CD assessment. MR enterography (MRE) offers radiation-free visualization of bowel-wall inflammation, thickening, strictures, fistulae, and extraintestinal complications , and CT enterography (CTE) is widely used in acute and resource-constrained settings where MRE is unavailable. Validated activity scores such as the Magnetic Resonance Index of Activity (MaRIA) quantify disease at a single time point but do not predict future outcomes . Even where a manual score has been linked to prognosis, the simplified MaRIA (sMaRIA) is only *associated* with future surgery at the group level in the PROFILE trial ; such work stops short of individualized, automated risk prediction. The prognostic value of

[1] Corresponding author: moti.freiman@runi.ac.il

imaging, its ability to predict what will happen to an individual patient, remains largely unexplored in an automated, quantitative framework, and almost all deep-learning work to date is single-modality.

Three gaps in particular motivate this study. First, there is **no large annotated CD enterography dataset**: expert per-finding annotation does not scale to thousands of studies, so most reported cohorts are small and single-institution. Second, **CT and MR are modeled separately**, doubling the engineering, validation, and deployment burden, and to our knowledge no published study trains a single deep model jointly on CTE and MRE for CD. Third, **classification is not prognosis**: most deep learning in CD addresses binary activity or scoring at a single time point , answering "is the patient sick now?" rather than the more actionable "will this patient need surgery, and when?". In addition, venue-grade evidence (multi-center external validation, calibration, and clinical net benefit) is frequently absent from CD imaging studies.

We address these gaps with a two-stage, modality-conditioned foundation-model pipeline and make four contributions. (i) **Automatic labeling at scale.** A clinical natural-language model, HSMP-BERT, extracts ten radiological findings from free-text Hebrew enterography reports and is applied identically to MRE and CTE, converting routinely acquired reports into structured labels without manual annotation. (ii) **A unified CT+MR detector.** A single frozen DINOv3 backbone with low-rank adapters, dual-branch multi-instance learning (MIL), and lightweight modality conditioning detects the ten findings from both modalities, matching the MRE specialist and outperforming the CTE specialist; we show that a modality token with feature-wise linear modulation (FiLM) is the sweet spot, while domain-specific normalization (DSBN) causes negative transfer on the dominant modality. (iii) **A two-stage detection-to-prognosis design.** The detected findings feed a penalized Cox model that issues individualized time-to-event predictions for surgery, steroid dependence, and biologic switch, jointly on CT and MR. (iv) **A rigorously clean CD-only cohort.** We exclude ulcerative colitis (UC), a distinct colon-only disease whose outcomes are not comparable to CD, and repair a patient-level data-leakage issue, yielding an honest CD-only evaluation with multi-center generalization, calibration, and decision-curve evidence.

Results

Our pipeline reads a CT or MR enterography study and predicts individualized outcomes in two stages: a unified detector that reports ten radiological findings, and a penalized survival model that fuses them with clinical features (Figure 1).

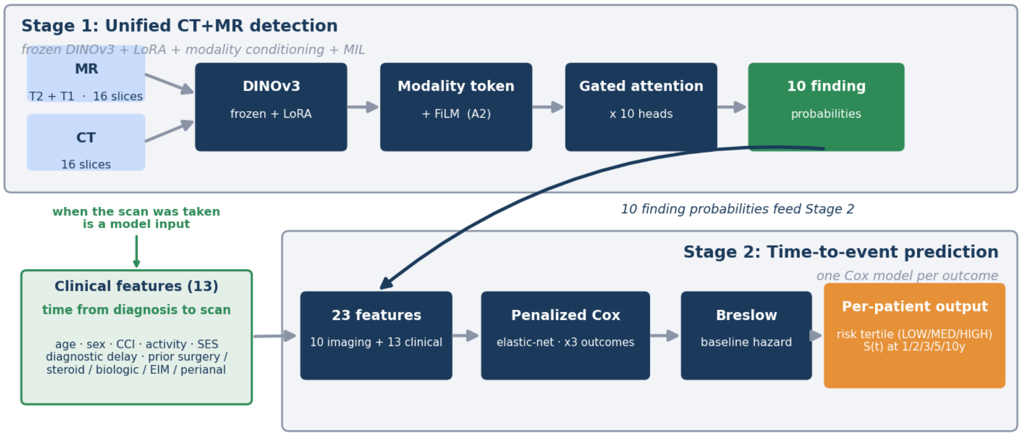

Figure 1 | Two-stage pipeline. *Stage 1 detects 10 radiological findings from CT and MR enterography with a frozen DINOv3 backbone, LoRA adapters, multi-instance learning, and modality conditioning (modality token and FiLM). Stage 2 fuses the 10 finding probabilities with 13 clinical features in a penalized Cox model to predict individualized time to surgery, steroid dependence, and biologic switch.*

Study cohort

From a full imaging asset of 8,053 enterography studies across four centers (Mor, Assuta, SZMC, and Rambam; MRE and CTE), we excluded 883 UC studies (510 MRE + 373 CTE). UC is a distinct, colon-only disease whose surgical outcome (colectomy) is not comparable to the small-bowel resection modeled here; the excluded UC studies show a textbook UC signature (low on every ileal finding, higher on colonic inflammation), confirming label validity. UC subtype was taken from the IBD registry and was 100% consistent across database versions. The resulting clean CD detection cohort comprises **7,170 studies** (5,713 MRE + 1,457 CTE), split at the patient level into 5,894 train / 623 validation / 653 test. A further 1,839 studies without a registry subtype (mostly healthy controls) were retained but flagged separately. The survival cohort (CD studies with linked outcomes) comprises **5,148 studies** (3,701 MRE + 1,447 CTE), with 782 surgery, 231 steroid-dependence, and 1,092 biologic-switch events. Table 1 summarizes the cohort by center and modality. A data-leakage issue in an earlier split (204 patients with studies spanning train/val/test) was repaired by majority reassignment, after which zero patients span splits.

Table 1 | Study cohort by disease group, stage, center, and modality. *The clean Crohn's-disease (CD) detection cohort excludes ulcerative colitis (UC) and is split at the patient level. The survival cohort is the subset of CD studies with linked time-to-event outcomes. All splits are leakage-checked (zero patients spanning splits).*

Cohort / partition	Studies	MRE	CTE	Notes
Full imaging asset	8,053	–	–	4 centers, MRE + CTE

Cohort / partition	Studies	MRE	CTE	Notes
Excluded: ulcerative colitis (UC)	883	510	373	distinct disease
No registry subtype (flagged)	1,839	≈1,831	≈8	mostly healthy controls
Clean CD detection cohort	**7,170**	**5,713**	**1,457**	
Train	5,894	–	–	patient-level
Validation	623	–	–	patient-level
Test (held out)	653	–	–	patient-level
Survival cohort (CD, with outcomes)	**5,148**	**3,701**	**1,447**	
Surgery events	782	–	–	
Steroid-dependence events	231	–	–	
Biologic-switch events	1,092	–	–	

Center-level composition of the CD cohort: Mor, Assuta, SZMC, and Rambam; all CTE examinations originate from a single center (Assuta), which is why leave-one-center-out validation is feasible on MRE only.

Automatic labeling from free-text reports

Rather than expert per-finding annotation, which does not scale to thousands of studies, we derived the ten radiological finding labels (Table 2) automatically from the free-text radiology reports using HSMP-BERT, a clinical natural-language model for Hebrew enterography text (Methods). The extractor was applied *identically* to MRE and CTE reports, so the same label definition governs both modalities, and both the unified detector and the modality specialists are evaluated against the same labels. This design makes the labeling step a reusable component of the pipeline rather than a one-off annotation effort, and it is what enabled a cohort two orders of magnitude larger than typical expert-annotated CD imaging studies. Because the labels are model-derived from report text rather than prospective human ground truth, all *absolute* performance figures inherit this label ceiling; comparisons between models are unaffected because they are scored against the same labels. As an internal consistency check, on the subset that also carried previously validated binary disease-activity labels the extracted findings reproduced those labels on all 6,223 studies. A prospective per-finding validation against radiologist annotation on a held-out subset is an important next step and a limitation of the present label set.

Stage 1: unified CT+MR finding detection

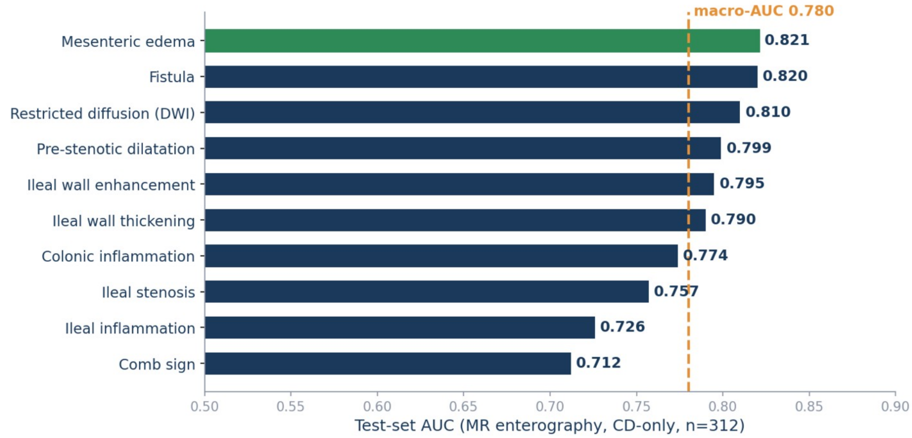

Figure 2 | Per-finding detection performance. *Held-out Crohn's-only MR test set (n=312); macro-AUC 0.780.*

On the held-out CD-only test set, the unified detector (configuration A2: frozen DINOv3-Base with low-rank adapters, dual-branch gated-attention MIL, and lightweight modality conditioning via a learned modality token plus FiLM) reached a macro-AUC of **0.780 (95% CI 0.750–0.810) on MRE** and **0.744 (0.691–0.789) on CTE**. Per-finding AUCs are reported in Table 2. Detection is strongest for the penetrating and stricturing phenotypes most relevant to prognosis (fistula 0.820, mesenteric edema 0.821, restricted diffusion 0.810, pre-stenotic dilatation 0.799, wall enhancement 0.795 on MRE) and weakest for comb sign (0.712).

Retraining Stage 1 end to end on the clean CD-only data barely moved validation performance (validation macro-AUC 0.8375, versus 0.8399 for the pre-exclusion model), confirming that excluding UC did not degrade CD detection. Among modality-conditioning components, a modality token and FiLM matched dedicated per-modality specialists, whereas DSBN over-parameterized this moderately imbalanced ($\approx 20\%$ CT) setting and reduced the dominant-modality (MRE) macro-AUC; we therefore adopted token+FiLM (A2) and omitted DSBN.

Table 2 | Per-finding detection AUC (95% CI) on the held-out CD-only test set *for the unified A2 model, reported separately for MR enterography (MRE, $n = 312$) and CT enterography (CTE, $n = 170$). Restricted diffusion (DWI) is not assessable on CTE (no DWI sequence). Macro-AUC is the unweighted mean of the per-finding AUCs. Confidence intervals are paired/cluster bootstrap by patient ($n = 1{,}000$).*

Finding	Tier	MRE AUC [95% CI]	CTE AUC [95% CI]
Ileal	1	0.726 [0.668,	0.721 [0.637,

Finding	Tier	MRE AUC [95% CI]	CTE AUC [95% CI]
inflammation		0.785]	0.797]
Ileal wall enhancement	1	0.795 [0.741, 0.844]	0.667 [0.582, 0.748]
Ileal wall thickening	1	0.790 [0.736, 0.840]	0.731 [0.640, 0.810]
Restricted diffusion (DWI)	1	0.810 [0.765, 0.855]	n/a
Ileal stenosis	1	0.757 [0.697, 0.811]	0.780 [0.700, 0.849]
Pre-stenotic dilatation	1	0.799 [0.734, 0.853]	0.802 [0.721, 0.867]
Comb sign	1	0.712 [0.628, 0.785]	0.777 [0.668, 0.873]
Fistula	2	0.820 [0.749, 0.882]	0.822 [0.577, 0.982]
Colonic inflammation	2	0.774 [0.679, 0.860]	0.783 [0.670, 0.876]
Mesenteric edema	2	0.821 [0.716, 0.915]	0.619 [0.455, 0.774]
Macro-AUC		**0.780 [0.750, 0.810]**	**0.744 [0.691, 0.789]**

The per-modality *n* (312 MRE, 170 CTE) counts the held-out test studies scored for finding detection; the 653-study test partition in Table 1 additionally includes studies without extractable per-finding labels (e.g. registry-unknown controls).

Stage 2: individualized time-to-event prediction

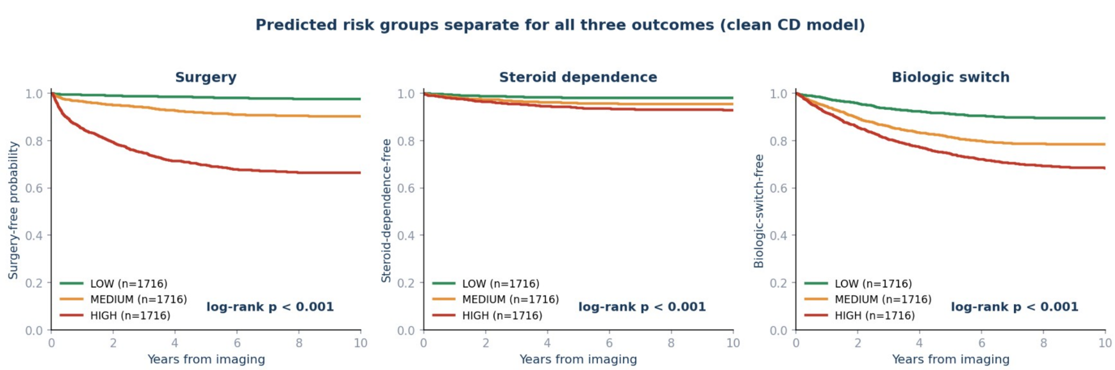

Figure 3 | Risk stratification for all three outcomes. *Kaplan-Meier event-free curves by predicted risk tertile (low / medium / high) on the unified CD cohort (n=5,148) for surgery,* *steroid dependence, and biologic switch. All three separate significantly (log-rank $p < 0.001$), with the sharpest separation for surgery and biologic switch.*

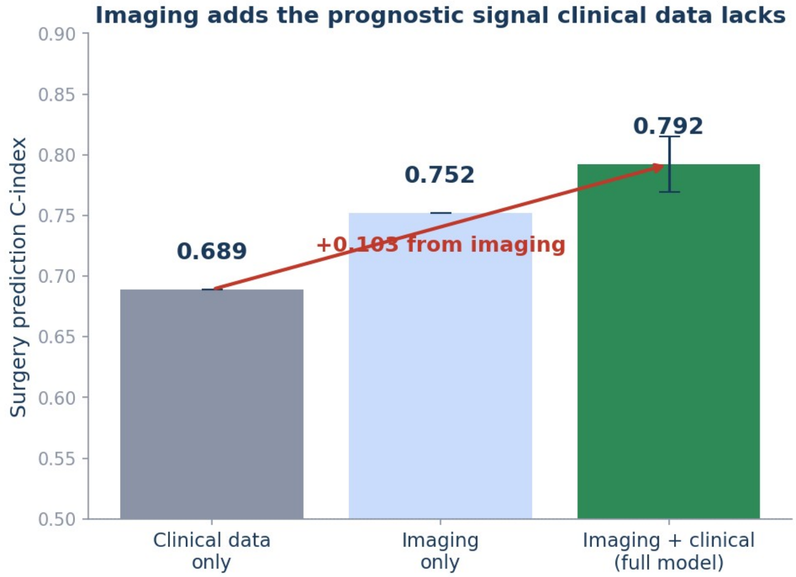

Figure 4 | Imaging adds prognostic signal. *Surgery C-index: clinical-only 0.689, imaging-only 0.752, combined 0.792 (imaging adds +0.103 over clinical).*

Using the A2 detector to generate imaging biomarkers, we fit an elastic-net penalized Cox model (l1-ratio 0.5) on the unified CD survival cohort (5,148 studies, 3,701 MRE + 1,447 CTE) with patient-grouped 5-fold cross-validation and Breslow baseline estimation. For **surgery**, the combined model (10 imaging findings + 13 clinical features) achieved a concordance index of **0.792 (95% CI 0.776–0.808)**, with a 5-fold mean of 0.794 ± 0.023 (Table 3). Imaging alone reached 0.752 and clinical alone 0.689, so imaging added ΔC-index$=+0.103$ over clinical features. On a per-modality basis, surgical signal was carried mostly by MRE (MR-only surgery C-index 0.796, $n = 3{,}701$), while CTE alone was weaker but non-trivial (CT-only 0.711, $n = 1{,}447$); the unified combined cohort reached 0.792. For steroid dependence and biologic switch, clinical history dominated and imaging added little (Table 3).

Table 3 | Time-to-event prediction on the unified CD CT+MR cohort *($n = 5{,}148$; patient-grouped 5-fold cross-validation, Breslow baseline). The pooled C-index is reported with a bootstrap ($n = 1{,}000$) 95% confidence interval, alongside the 5-fold mean±SD and the imaging-only / clinical-only / combined feature sets. ΔMRI = combined − clinical-only. Higher is better.*

Outcome	Events	Imaging	Clinical	Combined [95% CI] (±SD)	ΔMRI
Surgery	782	0.752	0.689	**0.792** [0.776–0.808] (± 0.023)	+0.103
Steroid dependenc	231	0.575	0.618	0.655 [0.619–	+0.037

Outcome	Events	Imaging	Clinical	Combined [95% CI] (±SD)	ΔMRI
e				0.692] (± 0.009)	
Biologic switch	1,092	0.586	0.604	0.643 [0.626–0.660] (± 0.021)	+0.039

Feature importance

Consistent with the surgery ablation, the imaging biomarkers corresponding to penetrating and stricturing phenotypes (fistula, stenosis, wall thickening, wall enhancement) carry the largest standardized surgical hazards, with the majority of elastic-net-selected features being imaging-derived. The clinical `prior surgery` indicator is strongly protective, reflecting that patients who already underwent resection are at lower risk of a *subsequent* surgical event within the follow-up window. The finalized per-feature hazard ratios are given in Table 4.

Table 4 | Hazard ratios for surgery *(per 1 SD, ridge-penalized Cox with robust standard errors, unified CD cohort, n=5,148). Imaging biomarkers are prefixed IMG. Clinical covariates not shown (age, comorbidity, socioeconomic status, diagnostic delay, time to scan, sex, prior EIM, prior perianal, prior biologic switch) had hazard ratios within 0.98–1.03 and p > 0.08.*

Feature	HR per 1 SD	95% CI	p
IMG: fistula	1.27	1.21–1.33	<0.001
IMG: stenosis	1.17	1.13–1.21	<0.001
IMG: wall thickening	1.16	1.14–1.19	<0.001
IMG: colonic inflammation	1.16	1.11–1.21	<0.001
IMG: wall enhancement	1.14	1.11–1.17	<0.001
IMG: comb sign	1.09	1.06–1.13	<0.001
IMG: restricted diffusion	1.07	1.04–1.11	<0.001
IMG: pre-stenotic dilatation	1.06	1.03–1.10	<0.001
Disease activity	1.06	1.02–1.10	0.004
Prior steroid dependence	1.06	1.01–1.10	0.013
Prior surgery	0.71	0.70–0.73	<0.001

External validation and clinical utility

Multi-center generalization (leave-one-center-out).

Leaving out each MRE center in turn and refitting the Cox model, the held-out-center surgery C-index was **0.754** ± **0.053** (Mor Inside 0.823, Assuta 0.694, SZMC 0.745; Table 5). CT-only leave-one-center-out is not feasible because all CTE examinations originate from a single center (Assuta); we therefore report MRE leave-one-center-out as the external-generalization proxy.

Table 5 | Leave-one-center-out (LOCO) external validation for surgery *on the MRE cohort (combined model). Each row leaves out one contributing center and evaluates on that held-out center. The mean ± SD across held-out centers is the external-generalization proxy; CTE is single-center and therefore not amenable to LOCO.*

Held-out center	Surgery C-index
Mor Inside	0.823
Assuta	0.694
SZMC	0.745
Mean ± **SD**	**0.754** ± **0.053**

Calibration.

The combined surgery model was well calibrated on the clean CD-only cohort, with an integrated Brier score of **0.074 (95% CI 0.068–0.079)** over the 0.1–0.9 event-time grid (lower is better).

Unified model versus dedicated specialists.

To test whether one conditioned model can replace two single-modality models, we trained a dedicated MRE specialist and a dedicated CTE specialist (single modality, no modality conditioning) on the same clean CD data and evaluated all three on the held-out CD test set. On MRE the unified model matched the specialist (macro-AUC 0.780 [0.750–0.810] vs. 0.793 [0.764–0.820]; overlapping intervals), and on CTE the unified model *outperformed* the specialist (0.744 [0.691–0.789] vs. 0.673 [0.606–0.729]). The gain on CTE is consistent with the far smaller CTE training set benefiting from joint cross-modality learning, whereas the data-rich MRE side is unchanged. A single unified model therefore replaces two specialists at no cost on MRE and with a clear gain on CTE.

Decision-curve analysis.

On the clean CD cohort the observed 5-year surgery incidence was 13.4% (Kaplan-Meier). Across the full range of clinically plausible decision thresholds (5% to 50% predicted 5-year risk), using the model to flag high-risk patients produced a positive net benefit that exceeded both the treat-all and the treat-none defaults at every threshold (for example, net benefit 0.078 at a 10% threshold versus 0.038 for treat-all, and 0.033 at a 25% threshold where treat-all is already net-harmful). Acting on the model's risk score is therefore clinically preferable to intervening on everyone or no one throughout the relevant range.

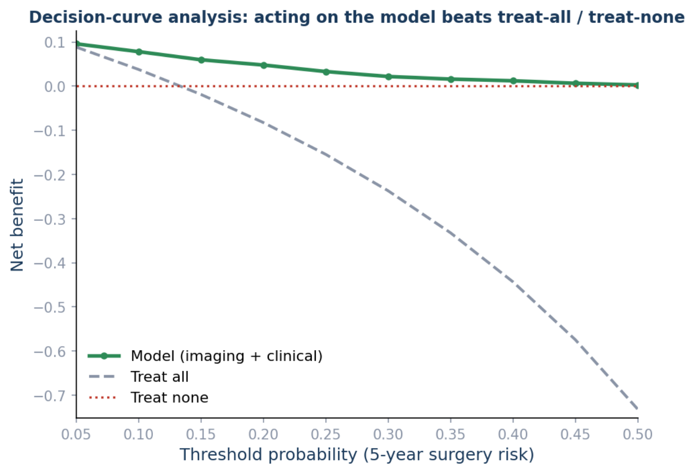

***Figure 5 | Decision-curve analysis** for the 5-year surgery horizon on the clean CD cohort. The model's net benefit (green) exceeds both treat-all (grey) and treat-none (red) across the full range of clinically plausible decision thresholds.*

Discussion

We asked whether one modality-conditioned foundation-model backbone, fed by automatically extracted labels, can serve both CT and MR enterography and deliver individualized CD prognosis. The answer is a qualified yes.

An honest, Crohn's-only accounting

The single most important framing point is that our headline detection figure is deliberately conservative. An earlier all-IBD evaluation reported a macro-AUC of 0.849, but that number was inflated by ulcerative-colitis easy-negatives: UC has essentially no ileal (small-bowel) disease, so UC studies are true-negatives on the ileal findings that dominate the macro-average, which artificially raises the score. Once UC is excluded and the model is retrained end to end on the clean CD-only data, the honest MRE macro-AUC is ≈ 0.780, essentially unchanged whether we evaluate the clean model or the previous model on the CD-only test set. We report the 0.780 figure and treat the 0.849 $\rightarrow$ 0.780 correction as a feature, not a defect: UC and CD are different diseases with non-comparable outcomes (colectomy vs. small-bowel resection), and mixing them both inflates apparent detection accuracy and contaminates a survival model whose surgical endpoint is CD-specific. Excluding UC barely changed validation performance (0.8375 vs. 0.8399), confirming that the clean cohort costs almost nothing in CD detection while making every downstream claim defensible.

Imaging adds prognostic signal for surgery; clinical history dominates the rest

For surgery, imaging contributes substantial prognostic value (ΔC-index $= +0.103$ over clinical-only), with penetrating and stricturing imaging phenotypes carrying the strongest hazard. A patient with radiological evidence of fistula, stenosis, or pre-stenotic dilatation faces elevated surgical risk even when clinical history appears unremarkable. The outcome-specific pattern is informative: steroid dependence is driven by treatment history (a clinical, not imaging, phenomenon), and biologic switch reflects both morphology and history, so imaging adds little to either. This argues against a single-modality view of CD prognosis and for reporting per-outcome, rather than aggregate, contributions.

Lightweight conditioning matches specialists; DSBN causes negative transfer

Among modality-conditioning components, a modality token and FiLM are nearly free and match dedicated specialists, whereas domain-specific normalization (DSBN) adds per-modality parameters that this moderate, imbalanced CT+MR scale cannot support, degrading the dominant modality (MRE). The practical recommendation is concrete: condition with a token and FiLM, and omit DSBN. This reconciles an apparent contradiction with the cross-modal literature, where DSBN helps in large-scale adaptation; at a moderate, imbalanced scale ($\approx 20\,\%$ CT) the heavier component flips from helpful to harmful. We frame unification as *parity* (no detected accuracy difference relative to two specialists) rather than superiority, since we did not run a formally powered non-inferiority trial; the deployment value, one model instead of two to train, validate, calibrate, and maintain, is real and does not depend on a superiority claim.

A unified CT+MR survival model

An updated outcomes release linked time-to-event labels to CTE studies, enabling, to our knowledge for the first time in CD, a Cox prognostic model fit jointly on CTE and MRE. Most of the surgical signal is carried by MRE (MR-only C-index 0.796), but CTE contributes a non-trivial surgical signal on its own subset (CT-only 0.711), and combining the two does not degrade overall performance (unified 0.792). Because all CTE originates from a single center, external CT generalization remains to be shown, but the joint cohort already surfaces per-modality effects that a MR-only analysis would miss.

Limitations

- **Labels are model-derived.** The ten findings come from a report-text extraction model (HSMP-BERT), not prospective human annotation; both the unified and specialist models are scored against the same labels, so the comparison is unaffected, but absolute figures inherit label noise.
- **CTE is single-center.** All CTE examinations originate from Assuta, so multi-center (leave-one-center-out) external validation is reported on MRE only.
- **Retrospective design and proportional-hazards assumption.** Selection biases cannot be fully excluded; time-varying effects were not modeled.
- **Decision-curve evidence pending.** The combined surgery model is well calibrated (integrated Brier score 0.074 [0.068–0.079]), but the decision-curve net-benefit analysis on the clean CD-only cohort, and a unified-vs-specialist Stage-2 comparison, are being finalized.

- **Per-study modeling.** Stage 2 operates at the study level under patient-level splits; per-patient temporal aggregation of multi-scan patients is left to future work.

Conclusion

A single conditioning-light foundation-model backbone, fed by automatically extracted report labels, can detect CD findings from both CTE and MRE and deliver individualized, calibrated prognosis. On a rigorously clean CD-only cohort, the unified detector reaches an MRE macro-AUC of 0.780 and the survival head predicts time to surgery with a C-index of 0.792, with imaging adding 0.103 over clinical features, generalizing across centers (LOCO 0.754), and remaining well calibrated (integrated Brier score 0.074). A modality token plus FiLM is the conditioning sweet spot; domain-specific normalization should be omitted. We present the 0.849 → 0.780 correction and the honest, CD-only accounting as strengthening, rather than weakening, the case for unified, conditioning-light enterography pipelines.

Methods

Study population and cohort definition

This retrospective study pooled CD examinations from four Israeli centers (Mor, Assuta, Shaare Zedek Medical Center [SZMC], and Rambam), spanning MRE and CTE. The study was approved by the institutional review board of the coordinating center, and the requirement for individual informed consent was waived for this retrospective analysis of de-identified data; original imaging and clinical databases were treated as read-only. **[TODO: Insert the Helsinki/IRB approval number(s) and inter-center data-sharing agreement references (clinical PI team, SZMC).]**

Ulcerative-colitis exclusion.

Because the surgical endpoint (small-bowel resection) is CD-specific and not comparable to the UC surgical endpoint (colectomy), we excluded all UC studies. Subtype was taken from the IBD registry (`IBD_subtype`) and verified to be 100% consistent across database versions on the shared accessions. The 883 excluded UC studies (510 MRE + 373 CTE) exhibited a textbook UC signature (low on every ileal finding, higher on colonic inflammation), confirming that the subtype labels are meaningful. Studies without a registry subtype (1,839, mostly healthy controls not in the IBD registry) were retained but flagged separately.

Data splitting and leakage control.

All splits were performed at the patient level (a patient appears in only one split), grouped by `patient_id`. A leakage issue in an earlier split (204 patients whose studies spanned train/val/test) was repaired by majority reassignment (214 studies moved), after which zero patients span splits. The same patient-grouped 5-fold assignment was used consistently for both stages.

Imaging preprocessing

Each study was reduced to **16 axial slices** by uniform `linspace` sampling and resized to 224×224. For MRE, T2-weighted and T1-weighted post-contrast sequences were used; CTE was routed through the T2 branch with the T1 branch zeroed via a missing-modality mask. Because CTE spans the whole abdomen and thorax, the bowel was localized before slice sampling: TotalSegmentator (fast mode) produced small-bowel and colon masks whose union bounding box (with a 10-voxel pad) cropped the volume, after which 16 slices were evenly sampled on the cropped volume. This mirrors the MRE bowel-crop pipeline, so localization comes from the crop rather than the slice sampler.

Automatic label extraction (HSMP-BERT)

Ten radiological findings (Table 2) were extracted from the free-text (Hebrew) enterography reports using HSMP-BERT, a clinical natural-language information-extraction model for CD report text, applied identically to MRE and CTE reports. Graded inflammation outputs were binarized at grade ≥ 1. The extractor produces one binary label per finding per study; the same label definition governs both modalities and both the unified detector and the modality specialists. On the subset with previously validated binary disease-activity labels, the extracted findings reproduced those labels on all 6,223 studies; a prospective per-finding validation against radiologist annotation is planned. **[TODO: Add the HSMP-BERT model citation.]**

Clinical features

Thirteen clinical features (6 continuous, 7 binary) were extracted from the clinical outcomes table, representing disease characteristics and treatment history at the time of imaging: time from diagnosis to imaging, age at diagnosis, Charlson Comorbidity Index, disease-activity cluster, socioeconomic status, and diagnostic delay; and prior surgery, prior steroid dependence, prior biologic switch, prior extraintestinal manifestations, prior perianal disease, sex, and a clinical-data-availability indicator. Missing values were imputed from the training set only (median for continuous, mode for binary), with a `has_clinical` indicator so the model can discount imputed values.

Stage 1: unified finding detection

A **frozen DINOv3-Base** backbone extracts per-slice features; **low-rank (LoRA) adapters** ($\approx 8.5\%$ of parameters trainable) adapt it cheaply and stably. A dual-branch design processes T2 and T1 (CTE uses the T2 branch with T1 zeroed). Ten *independent* gated-attention MIL heads, one per finding, pool slices into bag representations:

$$a_k^{(i)} = softmax\left(w^{(i)\top}\left(\tanh\left(V^{(i)} h_k\right) \odot \sigma\left(U^{(i)} h_k\right)\right)\right), z^{(i)} = \sum_{k=1}^{16} a_k^{(i)} h_k,$$

where h_k is the fused representation of slice k and i indexes the finding. Each bag representation is concatenated with clinical features and passed through a per-finding multilayer perceptron.

Modality conditioning.

The unified model conditions the shared backbone on modality using (a) a learned **modality token** indicating CTE vs. MRE, and (b) **FiLM** , per-channel scale and shift predicted from the modality token. We evaluated a conditioning ladder: token only, token+FiLM (adopted as the production A2 configuration), and token+FiLM+**DSBN** , the last of which applies modality-specific normalization. DSBN degraded the dominant modality (negative transfer) and was omitted. Modality specialists (one backbone per modality, no conditioning) provide the head-to-head baseline.

Warm-start and training.

All Stage-1 models (specialists and unified variants) were warm-started from the same prior MR-only DINOv3 checkpoint, applied symmetrically so that any initialization advantage is conferred equally. Models were trained on a single NVIDIA A6000 (48 GB) in bfloat16 with a tier-aware asymmetric loss for class imbalance, AdamW with differential learning rates, and early stopping on validation macro-AUC. The clean CD-only retrain reached its best validation macro-AUC (0.8375) at epoch 8.

Stage 2: time-to-event prediction

For each study, the trained A2 detector produces ten sigmoid-calibrated finding probabilities $\hat{p}_1, \ldots, \hat{p}_{10}$, concatenated with the 13 clinical features to form a 23-dimensional vector. A separate elastic-net penalized Cox model is fit per outcome:

$$h(t / x) = h_0(t) \exp(\beta^{\top} x),$$

with ℓ_1 / ℓ_2 ratio 0.5 and a Breslow estimator giving the baseline cumulative hazard and individualized survival curves $\hat{S}(t / x)$. The two-stage design is motivated by (i) the supervision advantage (Stage 1 sees far more labeled studies than there are survival events), (ii) interpretability of the intermediate biomarkers, and (iii) statistical efficiency: at these event counts a low-dimensional Cox model respects the events-per-variable constraint (surgery: 782 events / 23 features $\approx$ 34) where an end-to-end deep survival model would overfit.

Evaluation and statistics

We report the concordance index (C-index; the probability that the model assigns higher risk to the patient with the sooner event, 0.5 = chance) with bootstrap ($n = 1{,}000$) 95% confidence intervals, and 5-fold patient-grouped cross-validation means ± SD. Feature-set ablations compare combined (23-dimensional) vs. imaging-only (10-dimensional) vs. clinical-only (13-dimensional) inputs, with ΔMRI = combined − clinical-only. External generalization was assessed by leave-one-center-out (LOCO) on MRE. Calibration (integrated Brier score) and decision-curve analysis on the clean CD-only cohort are being finalized. Stage-1 model pairs (unified vs. specialist, and ablation rungs) were compared with paired bootstrap confidence intervals on test-set macro-AUC (the unweighted mean of the ten per-finding AUCs).

Data availability

The de-identified derived dataset (ten finding probabilities and clinical features per study; no images) underlying the survival and decision-curve analyses can be shared subject to the inter-center data-sharing agreements. Raw DICOM data cannot be shared because original imaging is governed by individual hospital data-use agreements. Requests should be directed to the corresponding author.

Code availability

The training, inference, and evaluation code (frozen backbone with LoRA adapters, modality-conditioning modules, MIL head, survival fitting, and calibration/decision-curve evaluation) will be released on a public GitHub repository upon acceptance, with the repository URL added at the proof stage. Trained checkpoints will be made available to non-commercial academic users on reasonable request.

99

Torres J, Mehandru S, Colombel JF, Peyrin-Biroulet L. Crohn's disease. *The Lancet*. 2017;389(10080):1741–1755.

Peyrin-Biroulet L, Loftus EV, Colombel JF, Sandborn WJ. The natural history of adult Crohn's disease in population-based cohorts. *Am J Gastroenterol*. 2010;105(2):289–297.

Rimola J, Rodriguez S, Garcia-Bosch O, et al. Magnetic resonance for assessment of disease activity and severity in ileocolonic Crohn's disease. *Gut*. 2009;58(8):1113–1120.

Rimola J, Ordas I, Rodriguez S, et al. Magnetic resonance imaging for evaluation of Crohn's disease: validation of parameters of severity and quantitative index of activity. *Inflammatory Bowel Diseases*. 2011;17(8):1759–1768.

Bhandari P, et al. Simplified Magnetic Resonance Index of Activity (sMaRIA) is associated with future surgery but correlates only modestly with endoscopic inflammation in Crohn's disease: an analysis of the PROFILE trial. *Journal of Crohn's and Colitis*. 2026 (in press).

Stable E, Arnone D, Engel H, et al. Artificial intelligence in inflammatory bowel disease: a systematic review of diagnostic and prognostic applications. *Journal of Crohn's and Colitis*. 2024;18(Supplement 1):i246–i247.

Siméoni O, Vo HV, Seitzer M, et al. DINOv3. *arXiv:2508.10104*. 2025.

Baharoon M, Qureshi W, Ouyang J, et al. Towards general-purpose medical image representations: evaluating DINOv2 on radiology benchmarks. *arXiv:2312.02366*. 2024.

Raidium. Curia: a multi-modal foundation model for radiology. *arXiv:2509.06830*. 2025.

HONeYBEE: scalable multimodal AI in oncology through foundation-model-driven embeddings. *npj Digital Medicine*. 2025. `https://doi.org/10.1038/s41746-025-02003-4`.

Hu EJ, Shen Y, Wallis P, et al. LoRA: low-rank adaptation of large language models. *arXiv:2106.09685*. 2021.

Perez E, Strub F, de Vries H, Dumoulin V, Courville A. FiLM: visual reasoning with a general conditioning layer. *Proceedings of the AAAI Conference on Artificial Intelligence*. 2018;32(1).

Chang WG, You T, Seo S, Kwak S, Han B. Domain-specific batch normalization for unsupervised domain adaptation. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2019;7354–7362.

Wasserthal J, Breit HC, Meyer MT, et al. TotalSegmentator: robust segmentation of 104 anatomic structures in CT images. *Radiology: Artificial Intelligence*. 2023;5(5):e230024.

Ilse M, Tomczak J, Welling M. Attention-based deep multiple instance learning. *Proceedings of the 35th International Conference on Machine Learning (ICML)*. 2018;2127–2136.

Campanella G, Hanna MG, Geneslaw L, et al. Clinical-grade computational pathology using weakly supervised deep learning on whole slide images. *Nature Medicine*. 2019;25(8):1301–1309.

Li B, Li Y, Eliceiri KW. Dual-stream multiple instance learning network for whole slide image classification with self-supervised contrastive learning. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2021;14318–14328.

Plattner C, Sturm G, et al. IBDome: an integrated molecular, histopathological, and clinical atlas of inflammatory bowel diseases. *bioRxiv*. 2025. doi:10.1101/2025.03.26.645544.

Katzman JL, Shaham U, Cloninger A, Bates J, Jiang T, Kluger Y. DeepSurv: personalized treatment recommender system using a Cox proportional hazards deep neural network. *BMC Medical Research Methodology*. 2018;18(1):24.

Lee C, Zame W, Yoon J, van der Schaar M. DeepHit: a deep learning approach to survival analysis with competing risks. *Proceedings of the AAAI Conference on Artificial Intelligence*. 2018;32(1).

Pölsterl S. scikit-survival: a library for time-to-event analysis built on top of scikit-learn. *Journal of Machine Learning Research*. 2020;21(212):1–6.

Vickers AJ, Elkin EB. Decision curve analysis: a novel method for evaluating prediction models. *Medical Decision Making*. 2006;26(6):565–574.

Silverberg MS, Satsangi J, Ahmad T, et al. Toward an integrated clinical, molecular and serological classification of inflammatory bowel disease. *Can J Gastroenterol*. 2005;19(Suppl A):5A–36A.

Chikontwe P, Luna M, Kang M, Hong KS, Ahn JH, Park SH. Dual attention multiple instance learning with unsupervised complementary loss for COVID-19 screening. *Medical Image Analysis*. 2021;72:102105.